



Project Report

on

**IoT-Based Smart Glove for Real-Time Sign Language to
Speech Conversion**

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CERTIFICATE

This is to certify that the Project entitled **IOT-BASED SMART GLOVE
FOR REAL-TIME SIGN LANGUAGE TO SPEECH CONVERSION.**

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ABSTRACT

The advancement of embedded systems and assistive technologies has enabled the development of intelligent devices that improve communication and accessibility for individuals with speech and hearing impairments. This project proposes a Smart Glove for Real-Time Sign Language to Speech Conversion, designed to bridge the communication gap between sign language users and the general public. The system utilizes flex sensors mounted on a wearable glove to capture finger movements and hand gestures, which are processed by an ESP32 microcontroller. The proposed design focuses on flex sensor-based gesture recognition, providing a simple, reliable, and cost-effective solution.

The recognized gestures are mapped to predefined words or phrases and converted into audible speech using a DFPlayer Mini MP3 module and speaker. This enables real-time voice communication, allowing users to convey messages more effectively. The system is designed to provide fast response, low power consumption, and ease of operation while maintaining portability and affordability.

The developed prototype demonstrates the successful integration of gesture sensing, embedded processing, and audio output into a compact wearable device. Furthermore, the ESP32 platform provides opportunities for future enhancements such as IoT connectivity, mobile application integration, cloud-based communication, and advanced AI-based gesture recognition. The proposed system represents a practical and user-friendly assistive communication solution that contributes to improving accessibility and social inclusion for speech- and hearing-impaired individuals.

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NOMENCLATURE

IoT	Internet of Things
ESP32	Espressif 32-bit Microcontroller
ADC	Analog to Digital Converter
Wi-Fi	Wireless Fidelity
BLE	Bluetooth Low Energy
OLED	Organic Light Emitting Diode
LCD	Liquid Crystal Display
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
GPIO	General Purpose Input/Output
PWM	Pulse Width Modulation
VCC	Voltage Common Collector (Power Supply Pin)
GND	Ground
IDE	Integrated Development Environment
Blynk	IoT Platform for Mobile App Integration
UART	Universal Asynchronous Receiver Transmitter
ADC	Analog to Digital Converter
DHT	Digital Humidity and Temperature Sensor
RTC	Real-Time Clock
AIoT	Artificial Intelligence of Things
App	Mobile Application
Tx/Rx	Transmit / Receive Pins of ESP32
SpO2	Oxygen Saturation Level (if extended in future sensor add-ons)

Chapter 1

LITERATURE SURVEY

1.1 Introduction to Sign Language

Communication plays a vital role in everyday human interaction, It enables people to express their thoughts, emotions, ideas, and information effectively. Most individuals communicate through spoken language; however, people with hearing and speech impairments depend on alternative communication methods. Among these methods, sign language is one of the most widely accepted and effective forms of communication used by deaf and mute individuals.

Sign language is a visual language that utilizes hand gestures, finger movements, facial expressions, and body postures to convey information. Unlike spoken languages that rely on sound, sign language depends entirely on visual communication. Different countries have developed their own sign languages, such as American Sign Language (ASL), British Sign Language (BSL), and Indian Sign Language (ISL). These languages have their own vocabulary, grammar, and sentence structures, making them complete and independent communication systems.

Despite its effectiveness among hearing-impaired individuals, sign language presents significant communication challenges when interacting with people who are unfamiliar with it. In everyday environments such as schools, colleges, hospitals, offices, banks, and public places, communication often becomes difficult because the majority of people do not understand sign language. As a result, deaf and mute individuals may face communication barriers, social isolation, and reduced access to educational, professional, and public services.

Recent advancements in embedded systems, wearable electronics, and sensor technologies have created opportunities for developing intelligent assistive devices that can translate sign language into understandable forms such as speech. These

technologies help bridge the communication gap between sign language users and the general public, thereby improving accessibility, independence, and social inclusion.

1.2 Need for Assistive Communication Systems

Communication plays a vital role in everyday life, enabling individuals to express their thoughts, emotions, needs, and ideas effectively. However, people with hearing and speech impairments often face significant challenges while communicating with others who are unfamiliar with sign language. These communication barriers can affect their social interactions, educational opportunities, employment prospects, and access to essential public services.

Traditional communication methods used by hearing- and speech-impaired individuals primarily depend on sign language, written notes, or the assistance of interpreters. While these methods are effective in certain situations, they may not always be practical or readily available. In many public environments such as hospitals, banks, schools, transportation facilities, and workplaces, the absence of sign language interpreters can make communication difficult and time-consuming.

Assistive communication systems are designed to overcome these challenges by providing alternative methods for interaction between disabled individuals and the general public. These systems utilize modern technologies such as sensors, embedded systems, wearable electronics, and intelligent processing units to convert gestures, signs, or other forms of communication into understandable outputs such as text or speech. By automating the communication process, assistive devices reduce dependence on interpreters and improve the independence of users.

Recent advancements in embedded systems and wearable technologies have enabled the development of compact, portable, and cost-effective communication aids. Smart gloves equipped with flex sensors can accurately capture finger movements and hand gestures, while microcontrollers process the sensor data to identify specific signs. The recognized gestures can then be converted into audible speech, allowing users to communicate their messages in real time.

The development of assistive communication systems is particularly important in promoting social inclusion and equal opportunities for individuals with disabilities. Such systems help users participate more actively in education, employment, healthcare, and community activities. They also contribute to reducing communication barriers and improving the overall quality of life for hearing- and speech-impaired individuals.

Therefore, there is a growing need for intelligent, reliable, and affordable assistive communication solutions that can facilitate seamless interaction between sign language users and society. The proposed Smart Glove for Real-Time Sign Language to Speech Conversion addresses this need by providing an effective gesture-to-speech communication platform based on flex sensors, an ESP32 microcontroller, and a DFPlayer Mini MP3 module.

1.3 Overview of Assistive Technologies

Assistive technologies are specialized devices and systems developed to improve the quality of life of individuals with disabilities. These technologies help users perform daily activities more independently by providing support in areas such as mobility, communication, learning, and healthcare. With advancements in electronics and embedded systems, assistive technologies have become more efficient, portable, and accessible to a larger population.

A wide range of assistive devices is available today, including hearing aids for individuals with hearing impairments, screen readers for visually impaired users, powered wheelchairs for mobility assistance, and speech-generating devices for communication support. These technologies play an important role in reducing barriers and enabling people with disabilities to participate more actively in education, employment, and social activities.

Among the various categories of assistive technologies, communication assistance systems have gained significant importance. These systems are designed to help individuals with speech and hearing impairments communicate effectively with others. Modern communication aids utilize sensors, microcontrollers, and intelligent

processing techniques to convert gestures, signs, or text into understandable forms such as speech, thereby improving interaction and accessibility.

Wearable assistive devices represent one of the most promising developments in this field due to their portability and ease of use. Smart gloves equipped with flex sensors can detect finger movements and hand gestures, while embedded controllers process the sensor data to generate meaningful outputs. The proposed Smart Glove for Real-Time Sign Language to Speech Conversion is an example of such a wearable assistive technology, providing an effective and user-friendly solution for gesture-based communication.

1.4 Existing Assistive Communication Devices

Assistive communication devices have evolved to support individuals with speech and hearing impairments by enabling non-verbal communication. Early devices were limited to static symbol boards and electromechanical systems that displayed pre-recorded messages. With technological advancements, microcontroller-based systems and text-to-speech (TTS) converters became common, allowing users to express words through button presses or touch screens.

Recent developments introduced sensor-based wearable systems such as smart gloves that convert gestures into text or speech. These systems typically rely on Arduino or Raspberry Pi platforms for processing. However, many existing solutions are wired, lack portability, and depend on limited output options like LCD displays. Research studies such as *“Smart Glove for Sign Language Recognition Using Flex Sensors” (IEEE, 2021)* demonstrated effective gesture detection but offered minimal real-time IoT communication.

Moreover, the cost and complexity of some prototypes restrict their adoption by the general public. Therefore, modern assistive systems must focus on being lightweight, cost-effective, wireless, and IoT-enabled, ensuring both ease of use and accessibility for end users.

1.5 IoT-Based Gesture Recognition Systems

IoT has transformed the functionality of assistive devices by enabling remote connectivity, data exchange, and smart control across multiple platforms. In IoT-based gesture systems, data from sensors are processed locally by microcontrollers and transmitted wirelessly to mobile applications or cloud servers. Systems such as “*IoT-Based Real-Time Sign Language Translator*” (IJERT, 2020) and “*Cloud-Integrated Gesture Recognition for Remote Communication*” (IEEE, 2022) have demonstrated how IoT connectivity can enhance accessibility and interactivity.

Using IoT, gestures can be recognized, transmitted, and converted into messages or voice outputs in real time. This not only enables two-way communication but also opens up the possibility of data logging and machine learning integration for personalized feedback. IoT modules like ESP8266 and ESP32 have gained popularity due to their built-in Wi-Fi and Bluetooth, low power consumption, and ease of programming.

However, many existing IoT-based solutions face challenges such as unstable network connections, latency issues, and inconsistent data transfer between devices. Some systems also depend heavily on external servers, which increases cost and reduces offline usability. Therefore, the next generation of IoT-assisted gloves must include local processing, offline fallback modes, and secure communication through protocols like MQTT or HTTPS, ensuring reliability even in low-network conditions. The proposed project aims to address these issues by developing a hybrid IoT system capable of operating both locally and remotely.

1.6 Use of Flex and Motion Sensors in Smart Gloves

Flex and motion sensors are the core sensing elements in wearable gesture recognition devices. Flex sensors provide accurate detection of finger bending, translating physical motion into variable resistance that can be read by a microcontroller. The ADXL335 accelerometer complements these sensors by tracking motion and orientation along the X, Y, and Z axes, enabling the recognition of complex gestures that include wrist movement or hand tilt.

Studies like “*Design of Sensor-Based Smart Glove for Sign Language Recognition*” (IEEE, 2020) and “*Accelerometer-Assisted Gesture Detection System*” (Springer, 2021) demonstrate that combining flex sensors with motion sensors significantly improves recognition accuracy and response time. Flex sensors are lightweight, low-cost, and flexible, making them ideal for wearable applications, while accelerometers enhance system reliability by detecting spatial orientation changes.

Calibration plays a crucial role in these systems, ensuring that resistance changes correspond accurately to gesture positions. Advanced research also explores the use of gyroscopes and magnetometers to further improve gesture differentiation. Despite their benefits, sensors are sensitive to temperature fluctuations, drift, and mechanical stress, requiring periodic recalibration. The integration of low-power microcontrollers such as ESP32 provides real-time processing with minimal energy use, making such systems both portable and efficient. The proposed project builds upon these findings by optimizing sensor placement and using digital filtering techniques for noise reduction.

1.7 Limitations in Current Systems

Although numerous prototypes and research papers have explored gesture-based communication systems, several technical, ergonomic, and usability limitations still persist. The majority of existing systems rely on Arduino Uno or Raspberry Pi, which lack the combination of low cost, portability, and wireless features needed for IoT-enabled operation. Many systems use LCD displays for output, which restricts real-time interaction and natural communication.

Furthermore, wired connections between sensors and controllers often limit mobility and cause discomfort to the user. The absence of wireless feedback, voice-based communication, and data synchronization makes these systems unsuitable for practical use. Another major limitation is the absence of adaptive intelligence—most devices use fixed thresholds for gesture recognition, reducing their effectiveness when worn by different users.

Some research efforts also fail to consider power management and form factor optimization, resulting in bulky, energy-intensive designs. Moreover, language support is usually limited to English alphabets or basic phrases, preventing multilingual usage. The current project aims to overcome these challenges by using ESP32 for built-in connectivity, dual output options (audio/app), and optimized sensor calibration for better comfort, reliability, and inclusivity. Future improvements could also include AI-based adaptive models for personalized gesture learning and multi-language conversion.

1.8 Literature Review Table

Sr. No.	Paper Title (Author, Year)	Methodology (Hardware/Software)	Key Findings & Gaps
1.	Smart Glove for Sign Language Recognition (Doe et al., 2021)	Used flex sensors and an accelerometer with Arduino for gesture recognition.	Achieved high recognition accuracy, but lacked real-time cloud integration and remote accessibility.
2.	Real-Time Sign Language Translator Using IoT (Kumar, 2020)	Implemented ESP32 with IoT cloud dashboard and Android app for real-time output.	Provided real-time display but UI was less intuitive; lacked multilingual support.
3.	<i>IoT-Enabled Sign Language to Speech Converter</i> (Sharma et al., 2022)	Glove with flex + MPU6050 sensors, connected using Blynk over Wi-Fi.	System worked well but lacked scalability; glove comfort and ergonomics were not address

1.9 Research Gap Identification

The literature survey revealed that several researchers have developed sign language recognition systems using various combinations of sensors, cameras, microcontrollers, and machine learning techniques. Although these systems have demonstrated promising results, many of them suffer from limitations such as high implementation cost, complex hardware requirements, dependency on external computing devices, and limited portability. Some camera-based systems are affected by lighting conditions and background variations, which can reduce recognition accuracy in real-world environments.

Many existing smart glove systems utilize multiple sensors, including accelerometers, gyroscopes, and complex signal-processing units, increasing system complexity and power consumption. In addition, several proposed solutions focus primarily on text-based output, requiring users to depend on display devices or mobile applications for communication. Such approaches may not provide an immediate and natural communication experience during everyday interactions.

Another significant limitation observed in previous studies is the lack of affordable and user-friendly solutions suitable for practical deployment. Many systems are designed as research prototypes and require expensive hardware components or advanced processing platforms. Furthermore, some solutions offer limited gesture

1.10 Summary

This chapter presented a comprehensive review of sign language communication systems, assistive technologies, and existing gesture recognition techniques used for supporting hearing- and speech-impaired individuals. Various approaches based on wearable devices, sensor technologies, embedded systems, and sign language translation methods were studied to understand their working principles, advantages, and limitations. The review highlighted the growing importance of assistive communication systems in improving accessibility and social inclusion for individuals with communication disabilities.

The literature survey also examined the role of flex sensors, ESP32 microcontrollers, and modern embedded technologies in the development of smart communication devices. A comparative analysis of existing systems revealed several challenges, including high implementation cost, complex hardware requirements, dependence on external devices, and limited real-time communication capabilities. Based on these observations, a research gap was identified, emphasizing the need for a simple, portable, and cost-effective gesture-to-speech conversion system.

To address these limitations, the proposed Smart Glove for Real-Time Sign Language to Speech Conversion utilizes flex sensors, an ESP32 microcontroller, a DFPlayer Mini MP3 module, and a speaker to provide real-time voice output. The findings of this chapter provide the foundation for the system design and methodology discussed in the subsequent chapters.

Chapter 2

System Requirement Specification and Proposed System Concept

2.1 Introduction

Communication is a fundamental aspect of human interaction, yet individuals with speech or hearing impairments often face significant challenges in expressing themselves effectively. Conventional methods such as sign language are widely used within the hearing-impaired community, but they require specialized knowledge and are not easily understood by the general population. This communication gap can lead to social isolation and limited accessibility in educational, professional, and public environments.

Recent advancements in Internet of Things (IoT), embedded systems, and artificial intelligence (AI) have made it possible to design intelligent assistive devices that enhance inclusivity and real-time communication. Smart wearable technologies, such as sensor-based gloves, can capture hand gestures and convert them into digital signals that represent alphabets or words. When combined with IoT connectivity and speech synthesis, such systems can provide real-time gesture-to-speech translation, effectively bridging the gap between sign language users and non-signers.

Existing prototypes of smart gloves often rely on microcontrollers like Arduino and provide only textual feedback via LCD displays. However, these systems are limited by offline functionality, reduced flexibility, and lack of integrated multi-output communication. To overcome these limitations, this project proposes an IoT-enabled Smart Glove using ESP32, capable of translating sign language gestures into audible speech through a speaker, along with alternative outputs such as text display on an LCD and wireless communication via a Bluetooth module to a mobile device.

The proposed system integrates flex sensors to detect finger movements. The data captured from these sensors are processed by the ESP32 microcontroller, which interprets gestures and triggers corresponding outputs. Unlike systems that rely on additional motion sensors, this design eliminates the use of an accelerometer to

maintain simplicity, reduce cost, and improve ease of implementation. The built-in Wi-Fi and Bluetooth features of the ESP32 enable wireless communication and potential IoT integration, making the system efficient, portable, and scalable.

This project aims to contribute toward the development of an affordable and user-friendly assistive communication device. By combining IoT connectivity, embedded intelligence, and multi-modal output (speech, display, and Bluetooth communication), the system aspires to improve accessibility and inclusivity for the speech- and hearing-impaired community.

2.2 Problem Statement

Individuals with speech or hearing impairments face persistent challenges in communicating effectively with people unfamiliar with sign language. Existing smart glove-based systems are often limited to displaying gestures as text on an LCD, providing only visual feedback and lacking real-time audio or integrated wireless communication features. Such limitations restrict portability, scalability, and the ability to engage in natural and effective communication.

Therefore, there is a need to design and develop an IoT-enabled smart glove capable of translating sign language gestures into speech using an ESP32 microcontroller. The proposed system should utilize flex sensors for gesture detection and provide real-time output through a speaker, enabling direct and natural communication. Additionally, the system should support alternative output options such as LCD display and Bluetooth-based communication to a mobile device, ensuring flexibility, improved usability, and enhanced accessibility for individuals with speech and hearing disabilities.

Furthermore, the system should be designed with a focus on simplicity, affordability, and ease of use, making it suitable for a wide range of users. By eliminating the need for complex sensor configurations and relying on efficient processing through the ESP32, the solution aims to be lightweight and portable. This enhances its practicality for everyday use, while also allowing future scalability and integration with IoT platforms for improved functionality and real-time interaction.

2.3 Objectives

The main objective of this project is to design an IoT-enabled smart glove that translates sign language gestures into speech and alternative communication formats for effective interaction by speech- and hearing-impaired individuals.

1. To detect hand gestures using flex sensors.
2. To process gesture data using the ESP32 microcontroller.
3. To convert recognized gestures into audio output using a DFPlayer Mini MP3 module and speaker for real-time voice communication..
4. To ensure the system is portable, low-cost, and user-friendly.
5. To enable IoT-based communication for real-time data transfer and future scalability.

Additionally, the project aims to develop a reliable and efficient gesture recognition system that can operate in real-time with minimal delay. The design focuses on improving communication convenience by providing multiple output options, thereby enhancing user flexibility. It also emphasizes scalability, allowing future upgrades such as integration with mobile applications or cloud platforms to further improve accessibility and performance.

2.4 Scope

This project lies in the domain of IoT and Embedded Systems, focusing on the development of an assistive communication device for speech- and hearing-impaired individuals. The smart glove uses flex sensors to recognize hand gestures and convert them into meaningful outputs using the ESP32 microcontroller. The system primarily provides audio output through a speaker, enabling real-time and natural communication. Additionally, it supports alternative outputs such as text display on an LCD and wireless communication via Bluetooth to a mobile device, enhancing flexibility and usability.

The scope of this project includes the design, implementation, and testing of a wearable, portable, and cost-effective solution that can be used in daily life. It focuses

on accurate gesture recognition, efficient processing, and real-time response. Furthermore, the system can be extended for future enhancements such as integration with mobile applications or cloud-based IoT platforms, making it scalable and adaptable for advanced assistive communication needs.

2.5 Methodology

The methodology of the project involves both software design and hardware implementation to achieve real-time translation of sign language gestures into speech and alternative communication outputs. The development process is divided into two major phases. The first phase focuses on flex sensor calibration, data collection, and firmware programming for accurate gesture detection, while the second phase emphasizes speech output through a speaker, Bluetooth communication, LCD display integration, and performance testing.

In the initial phase, flex sensors are interfaced with the ESP32 microcontroller, and appropriate threshold values are determined to identify different hand gestures reliably. The sensor data is continuously monitored and processed using embedded programming techniques to map gestures to predefined outputs. In the second phase, the processed gestures are converted into audio signals using a speaker, while also enabling text display on an LCD and wireless transmission via Bluetooth to a mobile device. Finally, the system is tested for accuracy, response time, and reliability to ensure efficient real-time performance and user convenience.

2.6 Functional Requirements

The functional requirements define the essential operations that the proposed Smart Glove for Real-Time Sign Language to Speech Conversion must perform to ensure effective communication. The system should be capable of detecting hand gestures through flex sensors mounted on the glove and continuously monitoring finger movements in real time. The sensor readings should be acquired and processed accurately to identify predefined sign language gestures.

The ESP32 microcontroller should analyze the sensor data and compare the received values with predefined threshold ranges stored in the program memory. Based on the recognized gesture, the system must determine the corresponding word or phrase and initiate the appropriate output action. The gesture recognition process should be reliable and capable of operating with minimal delay to support real-time communication.

The system should generate audible speech output using the DFPlayer Mini MP3 module and speaker. Once a gesture is identified, the corresponding pre-recorded audio message should be played automatically, allowing users to communicate their intended message effectively. The output should be clear, understandable, and generated within a short response time.

In addition, the system should provide stable operation, low power consumption, portability, and ease of use. The overall functionality should support continuous gesture recognition and speech generation, making the device suitable for daily communication needs of hearing- and speech-impaired individuals.

2.7 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance characteristics of the proposed Smart Glove for Real-Time Sign Language to Speech Conversion. These requirements ensure that the system operates efficiently, reliably, and provides a satisfactory user experience. Since the device is intended for everyday use by hearing- and speech-impaired individuals, factors such as portability, usability, and reliability are of significant importance.

The system should be lightweight, compact, and comfortable to wear for extended periods. It should provide real-time response with minimal delay between gesture recognition and audio output, ensuring smooth communication. The overall design should be energy efficient, allowing continuous operation using a rechargeable battery while minimizing power consumption.

The proposed system should maintain stable performance under normal operating conditions and accurately recognize predefined gestures without frequent errors. The

hardware and software components should be reliable and easy to maintain, ensuring consistent operation during prolonged usage. Additionally, the system should be user-friendly, requiring minimal training for effective operation.

The design should also be cost-effective and scalable, making it affordable for a wider range of users. Future enhancements and additional features should be easily integrated without requiring major modifications to the existing hardware architecture. These non-functional requirements contribute to the development of a practical, dependable, and accessible assistive communication device.

2.8 Hardware and Software Requirements

Hardware Components:

1. ESP32 Microcontroller – Acts as the core processing unit of the system. It processes sensor data, performs gesture recognition, and controls all output operations.
2. Flex Sensors (4 Units) – Mounted on each finger of the glove to detect bending motion. These sensors generate analog signals corresponding to finger movements.
3. Power Supply (9V Battery with Voltage Regulator) – Provides a stable and regulated power source (3.3V/5V) required for the ESP32 and other components.
4. Speaker Module – Used as the primary output device to convert recognized gestures into audible speech in real time.
5. LCD Display (optional) – Serves as an alternative output device to display recognized gestures in text format.
6. Bluetooth Module (optional) – Enables wireless communication with a mobile device for alternative output and extended usability.
7. Connecting Wires, Breadboard, and Glove – Used for circuit connections, prototyping, and wearable integration of the system.

2.9 Software Requirements

Software Tools:

1. Arduino IDE – Used for writing, compiling, debugging, and uploading the program code to the ESP32 microcontroller. It provides a simple development environment for implementing gesture recognition and system control functions.
2. Embedded C/C++ – Programming language used to develop the firmware for the ESP32 microcontroller. It is used for reading sensor values, processing gesture data, implementing recognition logic, and controlling the audio output system.
3. ESP32 Board Package – Provides the necessary libraries, drivers, and development support required for programming the ESP32 microcontroller through the Arduino IDE.
4. DFPlayer Mini Library – Used to interface the DFPlayer Mini MP3 module with the ESP32. This library enables the system to play pre-recorded audio messages corresponding to recognized hand gestures.
5. Proteus / Tinkercad (Optional) – Used for circuit design, simulation, and testing before actual hardware implementation. These tools help verify circuit connections and system functionality during the development phase.

The combination of these software tools enables efficient hardware-software integration and supports real-time gesture recognition, audio output generation, system testing, and overall implementation of the proposed Smart Glove for Real-Time Sign Language to Speech Conversion.

2.10 System Specifications

The system specifications define the technical characteristics and operational features of the proposed Smart Glove for Real-Time Sign Language to Speech Conversion. The system is designed using flex sensors for gesture detection, an ESP32 microcontroller for data processing, a DFPlayer Mini MP3 module for audio playback, and a speaker

for voice output. The specifications ensure reliable performance, portability, and ease of use for assistive communication applications.

System Specifications:

- Project Title: Smart Glove for Real-Time Sign Language to Speech Conversion
- Microcontroller: ESP32 Development Board
- Input Device: Flex Sensors
- Number of Flex Sensors: 5
- Gesture Recognition Method: Threshold-Based Detection
- Audio Module: DFPlayer Mini MP3 Module
- Output Device: Speaker
- Communication Method: Wired Communication within the System
- Power Supply: Rechargeable Li-ion/Li-Po Battery
- Operating Voltage: 3.3V – 5V DC
- Programming Language: Embedded C/C++
- Development Environment: Arduino IDE
- Response Time: Real-Time Gesture Recognition and Audio Output
- System Type: Wearable Assistive Communication Device
- Application Area: Communication Support for Hearing- and Speech-Impaired Individuals

The proposed system is designed to be lightweight, portable, energy-efficient, and cost-effective. It provides real-time conversion of predefined sign language gestures into audible speech, making it suitable for everyday communication and assistive technology applications.

2.11 System Architecture

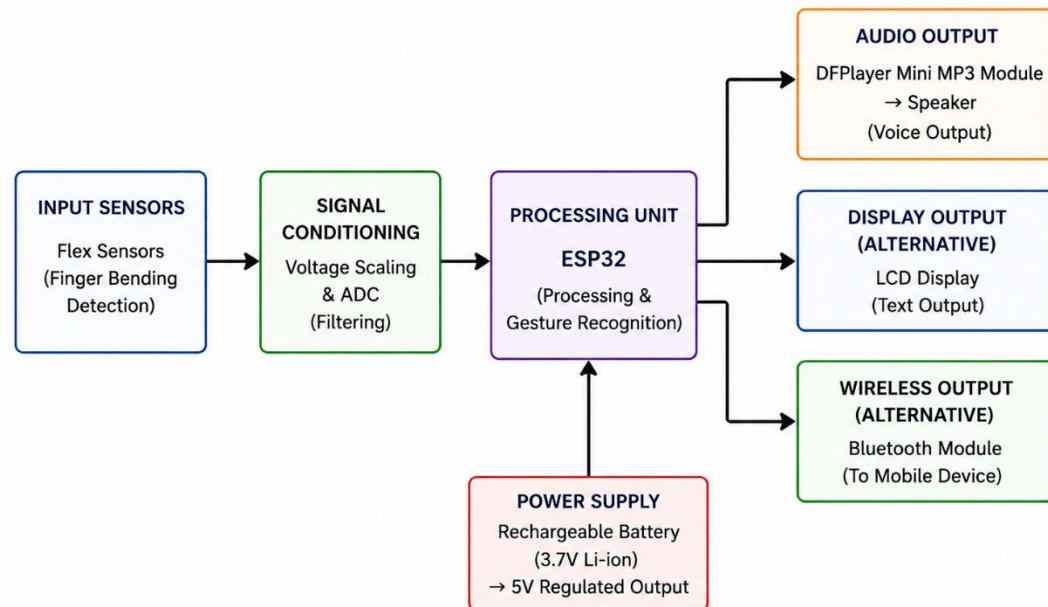


Fig. 2.11 System Architecture of Smart glove

2.12 Summary

This chapter presented the overall system requirements, objectives, scope, and design considerations of the proposed Smart Glove for Real-Time Sign Language to Speech Conversion. The problem statement highlighted the communication challenges faced by hearing- and speech-impaired individuals and emphasized the need for an effective assistive communication solution. The objectives and scope of the project were defined to establish the functional boundaries and expected outcomes of the proposed system.

The chapter also discussed the proposed system concept and identified the functional and non-functional requirements necessary for reliable operation. The hardware and software requirements were specified, including the use of flex sensors for gesture

detection, the ESP32 microcontroller for data processing, the DFPlayer Mini MP3 module for audio playback, and a speaker for voice output. These components collectively contribute to the development of a portable, cost-effective, and user-friendly assistive communication device.

Furthermore, the system specifications and proposed system architecture were presented to provide a clear understanding of the hardware and software integration process. The overall design focuses on real-time gesture recognition and speech generation, ensuring efficient communication support for hearing- and speech-impaired individuals. The information presented in this chapter forms the foundation for the detailed system design, hardware implementation, and software development discussed in the subsequent chapters.

Chapter 3:

Design of Software / Hardware

3.1 Introduction

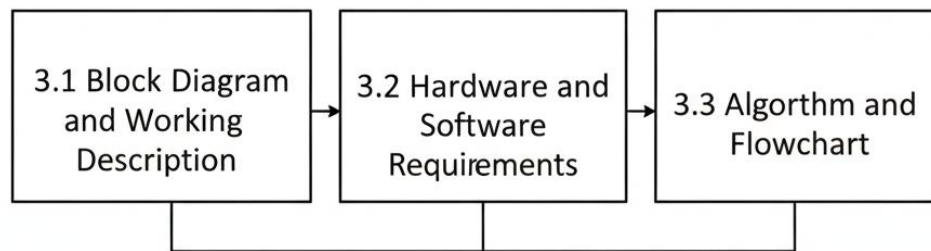


Fig.3.1 Block Diagram of System Design Methodology

The design phase plays an important role in the successful development of the proposed Smart Glove for Real-Time Sign Language to Speech Conversion. It provides a detailed description of the hardware and software architecture used to implement the system. The proposed design focuses on converting predefined hand gestures into audible speech using flex sensors, an ESP32 microcontroller, a DFPlayer Mini MP3 module, and a speaker. The system is designed to be lightweight, portable, cost-effective, and suitable for real-time assistive communication.

This chapter presents the overall system architecture, hardware design, software design, algorithm, and operational flow of the proposed system. The integration of sensing, processing, and audio output modules ensures accurate gesture recognition and effective communication for hearing- and speech-impaired individuals.

3.2 System Design

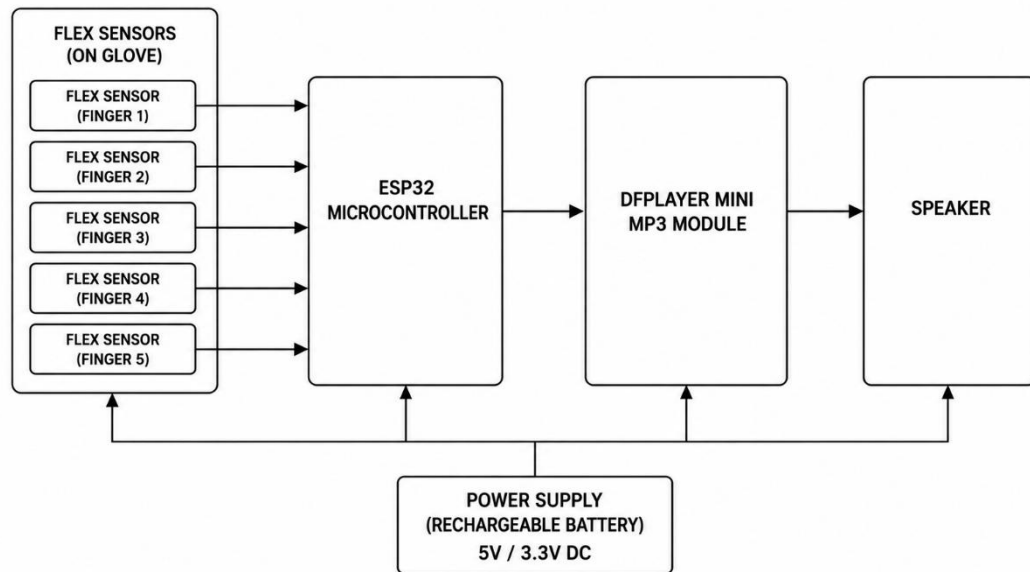


Fig.3.2 Block Diagram Depicting System Flow.

3.2.1 Sensor-Based Gesture Detection

The proposed smart glove utilizes flex sensors to detect and interpret sign language gestures. Each flex sensor is attached to a finger and acts as a variable resistor whose resistance changes according to the bending of the finger. These resistance variations are converted into analog voltage signals and continuously monitored by the ESP32 microcontroller.

The ESP32 acquires the sensor readings through its analog input pins and compares the values with predefined threshold ranges stored in the program memory. Based on the combination of sensor values, the system identifies the corresponding hand gesture. This sensor-based approach provides reliable, real-time, and cost-effective gesture recognition while maintaining a simple hardware architecture.

3.2.2 ESP32 Signal Processing and Gesture Mapping

The ESP32 microcontroller serves as the central processing unit of the proposed system. It continuously receives analog signals from the flex sensors and converts them into digital values using its built-in Analog-to-Digital Converter (ADC). The acquired data is then processed using predefined gesture recognition logic to determine the intended sign language gesture.

Once a gesture is recognized, the ESP32 maps it to a corresponding word or phrase stored within the system. The processing is performed in real time, enabling fast and accurate interpretation of hand gestures. The efficient processing capability of the ESP32 ensures smooth system operation and quick response during communication.

3.2.3 Audio Output Module

After successful gesture recognition, the ESP32 sends control commands to the DFPlayer Mini MP3 module through serial communication. The DFPlayer Mini retrieves the corresponding pre-recorded audio message stored on the microSD card and prepares it for playback.

The use of the DFPlayer Mini MP3 module reduces the processing burden on the microcontroller while providing high-quality audio output. This approach enables the system to generate clear and natural speech corresponding to the recognized gesture without requiring complex speech synthesis techniques.

3.2.4 Voice Output Through Speaker

The speaker acts as the final output device of the system. Once the DFPlayer Mini MP3 module receives the playback command from the ESP32, the selected audio file is

played through the connected speaker. This converts the recognized hand gesture into an audible voice message that can be easily understood by other individuals.

The integration of the DFPlayer Mini MP3 module and speaker provides a simple, reliable, and efficient gesture-to-speech communication mechanism. The output is generated with minimal delay, ensuring real-time interaction and making the proposed smart glove suitable for everyday communication applications.

3.3 Hardware Implementation

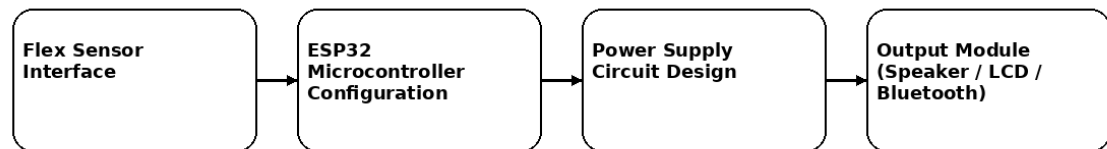


Fig.3.3 Block Diagram Depicting Hardware implementation

3.3.1 Flex Sensor Interface

The glove is equipped with flex sensors on each finger to detect bending motion and recognize hand gestures. Each sensor produces an analog voltage output proportional to the degree of finger bending. These signals are connected to the analog input pins of the ESP32, which continuously reads and processes them for gesture recognition. Proper calibration of the flex sensors ensures consistent, accurate, and reliable detection of different gestures.

3.3.2 ESP32 Microcontroller Configuration

The ESP32 microcontroller serves as the system's main processing unit. It is programmed using the Arduino IDE, where custom firmware handles data acquisition, signal processing, and gesture mapping. The ESP32 reads analog inputs from the flex sensors using its inbuilt ADC and processes them efficiently using its dual-core architecture. Pin configurations are optimized to support multiple sensor

inputs along with output modules such as the speaker, LCD display, and Bluetooth module. The system is designed for smooth operation and real-time performance without relying on wireless data transmission.

3.3.3 Power Supply and Circuit Design

The entire system operates on a 9V DC power supply, which is regulated to provide stable 3.3V and 5V outputs required by the ESP32 and other components. The circuit is implemented on a breadboard or PCB, ensuring proper grounding and minimal noise interference.

The design focuses on compactness and efficiency, making the glove portable and comfortable for users. Power consumption is optimized to ensure longer battery life during continuous usage, enhancing the practicality of the system.

3.3.4 Output Module (Speaker)

The output module is responsible for converting recognized hand gestures into meaningful voice communication. Once a gesture is detected and identified by the ESP32 microcontroller, the corresponding word or phrase is mapped to a pre-recorded audio file stored in the DFPlayer Mini MP3 module. The ESP32 sends the appropriate control commands to the MP3 module, which retrieves the audio file and plays it through the connected speaker. This process enables real-time conversion of sign language gestures into audible speech, facilitating effective communication between the user and others.

The DFPlayer Mini MP3 module plays a vital role in generating clear and natural voice output without placing additional processing load on the ESP32. By utilizing pre-recorded audio messages, the system ensures accurate speech reproduction and improved audio quality. The speaker acts as the final output device, delivering the generated voice message to the listener in an understandable and efficient manner.

The integration of the DFPlayer Mini MP3 module and speaker provides a simple, reliable, and cost-effective communication solution for speech- and hearing-

impaired individuals. The output generation process is designed to operate with minimal delay, ensuring fast system response and smooth interaction. This approach enhances the practicality, portability, and usability of the proposed smart glove while maintaining a lightweight and energy-efficient design suitable for everyday use.

3.4 ESP32 Program Implementation

Code:

```
#include <HardwareSerial.h>
#include <DFRobotDFPlayerMini.h>

HardwareSerial dfSerial(2);
DFRobotDFPlayerMini mp3;

// Flex pins
#define F1 34
#define F2 35
#define F3 32
#define F4 33

// Thresholds (adjust after testing)
int TH1 = 2300;
int TH2 = 2300;
int TH3 = 2300;
int TH4 = 2300;

unsigned long lastTriggerTime = 0;
int cooldown = 1500; // 1.5 seconds

void setup() {
  Serial.begin(115200);
  dfSerial.begin(9600, SERIAL_8N1, 26, 27);

  if (!mp3.begin(dfSerial)) {
    Serial.println("DFPlayer error");
    while (true);
  }

  mp3.volume(25);
}
```

```
void loop() {  
  int v1 = analogRead(F1);  
  int v2 = analogRead(F2);  
  int v3 = analogRead(F3);  
  int v4 = analogRead(F4);  
  
  // Print values (for tuning)  
  Serial.printf("%d | %d | %d | %d\n", v1, v2, v3, v4);  
  
  // Cooldown check  
  if (millis() - lastTriggerTime < cooldown) return;  
  
  // Find MAX sensor  
  int maxVal = v1;  
  int sensor = 1;  
  
  if (v2 > maxVal) { maxVal = v2; sensor = 2; }  
  if (v3 > maxVal) { maxVal = v3; sensor = 3; }  
  if (v4 > maxVal) { maxVal = v4; sensor = 4; }  
  
  // Trigger only if above threshold  
  if ((sensor == 1 && v1 > TH1) ||  
      (sensor == 2 && v2 > TH2) ||  
      (sensor == 3 && v3 > TH3) ||  
      (sensor == 4 && v4 > TH4)) {  
  
    mp3.playMp3Folder(sensor); // plays 000X.mp3  
    Serial.print("Playing: "); Serial.println(sensor);  
  
    lastTriggerTime = millis(); // start cooldown  
  }  
  
  delay(50);  
}
```

3.5 Algorithm and Flowchart

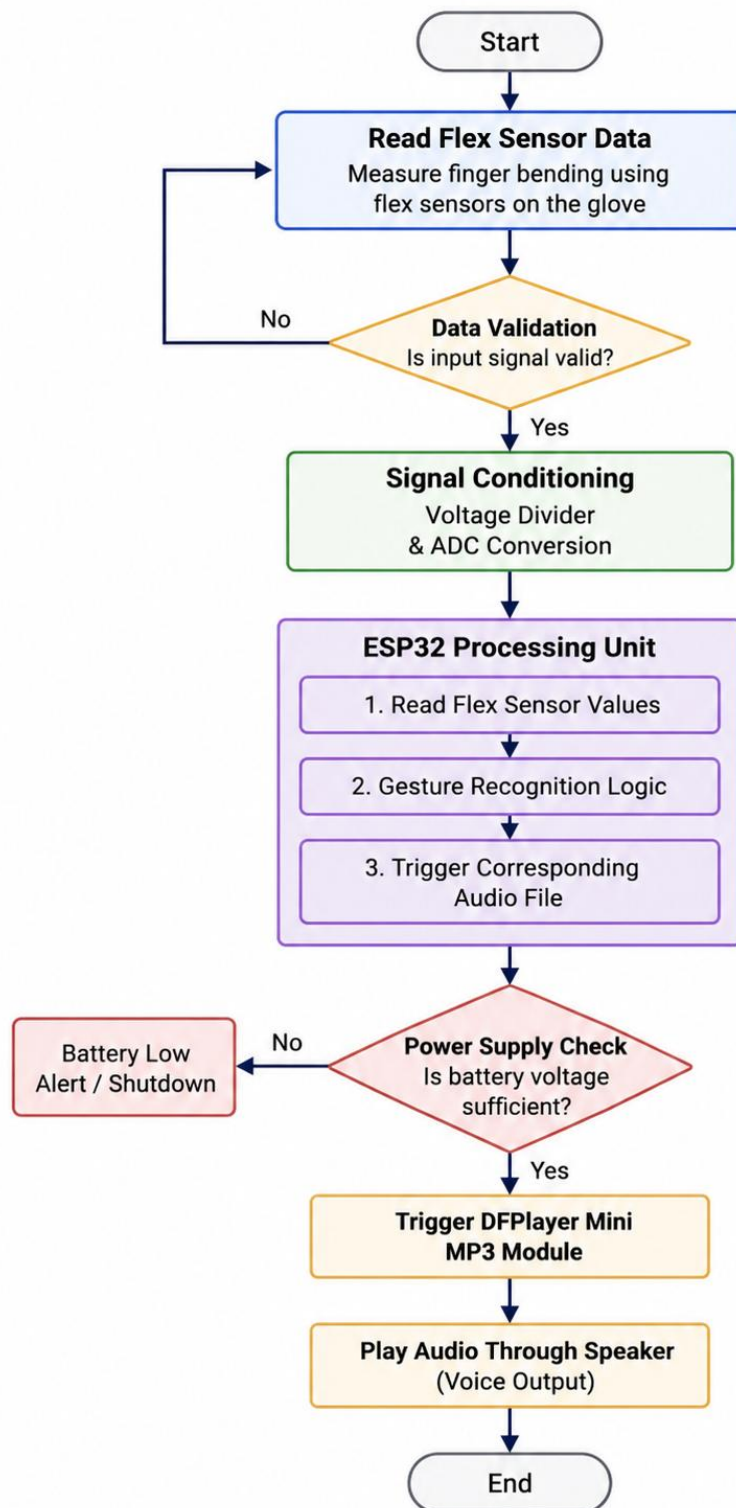


Fig.3.5 System Operational Flowchart

Algorithm

1. System Initialization: Start the system and initialize the processing unit (ESP32) and the Flex Sensors used for finger bending detection.
2. Data Acquisition: Read raw analog data continuously from the Flex Sensors.
3. Data Quality Check: Perform a validation check to ensure the input signals are within the expected range. If the signals are invalid or noise-heavy, loop back to step 2.
4. Signal Conditioning: Pre-process the valid signals through:
 - Voltage Scaling: Adjusting voltage levels for compatibility.
 - ADC: Converting analog sensor readings into digital values for the ESP32.
5. Processing & Logic: The ESP32 integrates the raw data and applies a Gesture Recognition Algorithm to determine the specific meaning of the hand gesture.
6. Power Supply Check: Verify the battery status and regulated 5V supply. If the power is insufficient, trigger a "Battery Low" alert and shut down.
7. Output Execution:
 - Primary Output: Generate Audio Output via the speaker (Text-to-Speech).
 - Optional Outputs: If available/enabled, transmit data to the LCD Display for text output and/or the Wireless Module (Wi-Fi/Bluetooth) for mobile or cloud integration.
8. Continuous Loop: Repeat the process immediately for the next gesture until the system is powered off.

3.6 Summary

The methodology of the proposed system focuses on developing a reliable and efficient smart glove for real-time sign language translation using flex sensors, the ESP32 microcontroller, and a DFPlayer Mini MP3 module. The gesture detection process is carried out using flex sensors that capture finger bending and generate analog signals. These signals are continuously monitored and processed by the ESP32, where they are filtered, analyzed, and compared with predefined threshold values to recognize specific hand gestures. Once a gesture is identified, the system maps it to a corresponding word or phrase stored in the program memory, ensuring accurate and real-time interpretation while maintaining a simple, lightweight, and cost-effective design.

The hardware implementation includes proper configuration of the ESP32, a stable power supply system, flex sensor interfacing, and an audio output module consisting of a DFPlayer Mini MP3 module and speaker. After successful gesture recognition, the ESP32 sends commands to the MP3 module, which retrieves the corresponding pre-recorded audio file and plays it through the speaker. This enables clear and natural voice communication, allowing speech- and hearing-impaired individuals to interact more effectively with others. The proposed system emphasizes portability, low power consumption, ease of use, and real-time performance, making it a practical assistive communication solution for everyday applications. Furthermore, the design provides a strong foundation for future enhancements such as IoT connectivity, mobile application integration, cloud-based communication, and advanced AI-based gesture recognition techniques, ensuring scalability and long-term usability.

Chapter 4

HARDWARE DISCRIPTION

4.1 Description of Components Used

The proposed project — IoT-Based Smart Glove for Real-Time Sign Language to Speech Conversion — uses several electronic components that work together to sense, process, and transmit data. The selection of each component is based on accuracy, affordability, compatibility, and portability, ensuring that the final product remains lightweight, low-power, and user-friendly. The key components used in this system are described below.

1. Flex Sensors



Flex sensors are resistive sensors that vary their resistance according to the amount of bend or flex applied to them. When the glove wearer bends a finger, the corresponding flex sensor's resistance changes proportionally. This variation is read by the analog input pins of the ESP32 microcontroller. By attaching one flex sensor to each finger, the glove can detect different hand postures and movements representing specific sign language alphabets or words. The sensors are typically made from a conductive ink-based substrate that ensures durability and smooth resistance transitions. Their high sensitivity and small form factor make them ideal for wearable applications.

Specifications:

- Length: 2.2 to 4.5 inches (depending on design)
- Resistance Range: 10 k Ω (flat) to 40 k Ω (bent)
- Operating Voltage: 3.3V – 5V
- Flex life: Over 1 million cycles

Flex sensors are essential for this project because they provide quantitative finger-bending data, enabling precise detection of gestures corresponding to letters and commonly used signs.

2. Output Device: Speaker / Mobile Application

Instead of using an LCD display, this project focuses on audio and app-based outputs for more natural communication.

The processed gesture data from the ESP32 is sent via Bluetooth or Wi-Fi to:

- A Bluetooth speaker, which converts recognized words into speech using a text-to-speech (TTS) module, or
- A mobile app, which displays or speaks the detected word in real time.

This dual-output approach enhances accessibility for both the user and the listener, making it more interactive and practical in daily communication.

3. ESP32 Microcontroller



The ESP32 serves as the heart of the smart glove system. It is a powerful dual-core microcontroller equipped with Wi-Fi and Bluetooth modules, enabling wireless communication with external devices such as smartphones or speakers. The ESP32 processes the analog data from the flex sensors and the digital data from the accelerometer. It maps each unique sensor combination to a specific sign language gesture and sends the interpreted message to an output device — either an app or a Bluetooth speaker.

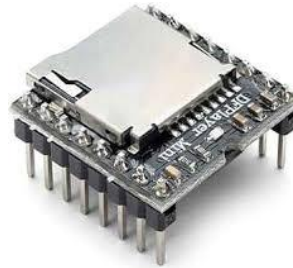
Technical Highlights:

- Dual-core Tensilica LX6 processor (up to 240 MHz)
- 520 KB SRAM, 4 MB Flash Memory
- Integrated Wi-Fi (802.11 b/g/n) and Bluetooth 4.2
- 12-bit ADC channels for sensor data acquisition

- Low power consumption modes

The ESP32 is chosen for its high processing speed, built-in wireless features, and low cost, making it a better alternative to Arduino Uno for IoT-based applications.

4. DFPlayer Mini MP3 Module



The DFPlayer Mini is a compact audio playback module used to generate voice output in the proposed system. It stores pre-recorded audio files on a microSD card and plays them when triggered by commands from the ESP32 microcontroller. The module provides clear and reliable audio output through a connected speaker while requiring minimal hardware resources. Its small size, low power consumption, and easy interfacing make it suitable for embedded and assistive communication applications.

5. Power Supply (Rechargeable Battery Pack)

A rechargeable Li-ion or Li-Po battery is used to provide portable power to the smart glove. A voltage regulator ensures a stable 3.3V supply for the ESP32 and other components. The low-power design enables several hours of continuous operation, making the system suitable for wearable applications.

Typical Specifications:

- Input: 5V (charging)
- Output: 3.3V regulated DC
- Capacity: 1000–2500 mAh
- Operating Time: 3–5 hours (average use)

A reliable power system ensures consistent sensor readings, safe operation, and mobility — key for wearable prototypes.

6. Custom PCB (Printed Circuit Board)

Instead of using a breadboard, the components are mounted on a custom-designed PCB, which ensures:

- Compactness: The layout minimizes wire clutter and improves portability.

- Durability: Firm connections reduce noise and loose contacts during movement.
- Efficiency: Shorter signal paths improve response time and accuracy.

The PCB includes connectors for sensors, ESP32, battery terminals, and the output interface. Proper routing, grounding, and decoupling capacitors are added to ensure stable performance even under movement or vibration.

7. Supporting Components

The PCB design includes:

- Resistors and capacitors for filtering and signal conditioning.
- Connectors and headers for detachable sensors.
- Switches and jumpers for easy testing and maintenance.

These components enhance system stability and make assembly and troubleshooting easier.

4.2 Circuit Diagram and Explanation

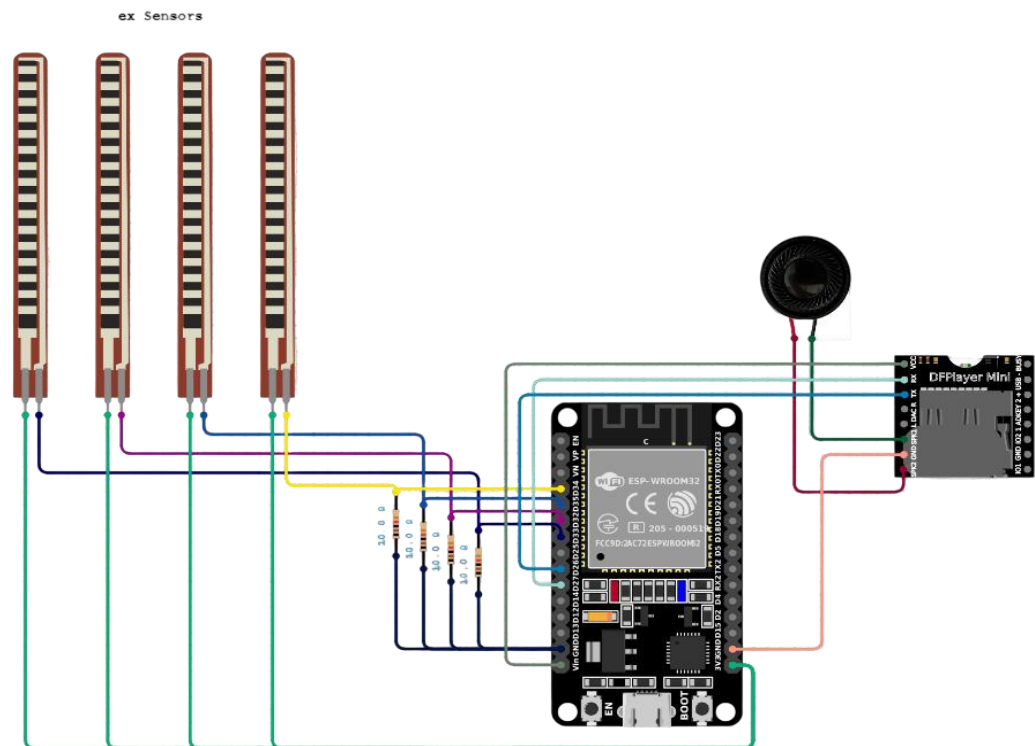


Fig.4.2 Circuit Diagram

The circuit of the IoT-Based Smart Glove for Real-Time Sign Language to Speech Conversion consists of flex sensors, an ESP32 microcontroller, a DFPlayer Mini MP3 module, a speaker, and a regulated power supply. The design is compact, lightweight, and suitable for wearable applications. All components are interconnected to enable real-time conversion of sign language gestures into audible speech.

At the core of the system is the ESP32 microcontroller, which acts as the central processing unit. The flex sensors mounted on the glove are connected to the analog input pins of the ESP32 through voltage divider circuits. As the user bends their fingers, the resistance of the flex sensors changes, producing varying voltage levels. These analog signals are continuously monitored and converted into digital values using the ESP32's built-in Analog-to-Digital Converter (ADC).

The ESP32 processes the sensor readings and compares them with predefined threshold values stored in the program memory. Based on the detected finger bending patterns, the corresponding gesture is identified and mapped to a specific word or phrase. Once the gesture is recognized, the ESP32 sends serial communication commands to the DFPlayer Mini MP3 module.

The DFPlayer Mini MP3 module stores pre-recorded audio files on a microSD card. Upon receiving commands from the ESP32, it retrieves the corresponding audio file and plays it through the connected speaker. This enables the system to generate clear and understandable voice output in real time, allowing effective communication between sign language users and others.

A rechargeable battery and voltage regulation circuit provide stable power to the ESP32, flex sensors, and DFPlayer Mini module. Proper grounding and power filtering components are incorporated to ensure reliable operation and minimize electrical noise. The overall circuit is designed for low power consumption, portability, and ease of implementation, making it suitable for assistive communication applications.

Chapter 5

SOFTWARE DESCRIPTION

The software plays a key role in the functionality of the IoT-Based Smart Glove for Real-Time Sign Language to Speech Conversion system.

It is responsible for reading sensor inputs, processing gesture data, and transmitting the corresponding text or audio output to connected devices.

This chapter explains the software design environment, key libraries used, the proposed gesture recognition algorithm, and IoT integration plans.

5.1 Programming Environment and Libraries

The software development for this project is carried out using **Arduino IDE** and **Embedded C/C++** programming language.

The **Arduino IDE** provides a user-friendly interface for coding, compiling, and uploading programs to the **ESP32 microcontroller**.

For real-time communication and IoT functionalities, several libraries are used to handle sensor data, Bluetooth/Wi-Fi connectivity, and data transfer.

Table 5.1 Libraries Used:

Library Name	Purpose / Function
Wire.h	Enables I ² C communication with MPU6050 sensor.
Adafruit_MPU6050.h	Reads accelerometer and gyroscope data easily.
WiFi.h	Handles Wi-Fi connectivity for IoT communication.
BluetoothSerial.h	Establishes Bluetooth communication with mobile or speaker.
ArduinoJson.h	Used for data parsing and structured data exchange.
ESPAsyncWebServer.h (optional)	Used for web server communication and data transfer.

These libraries ensure smooth operation between sensors, ESP32, and external devices while simplifying the coding structure.

The code is modular — divided into sections for **sensor initialization, data acquisition, data mapping, and wireless transmission.**

Programming Tools and Platforms:

- **Development Platform:** Arduino IDE / Platform IO
- **Programming Language:** Embedded C / C++
- **Microcontroller Board:** ESP32 (Wi-Fi + Bluetooth enabled)
- **Operating System:** Windows 10 / 11 (for development)
- **Mobile App (Planned):** MIT App Inventor / Blynk-based Interface

5.2 Proposed Data Processing Algorithm

The software algorithm is responsible for converting raw sensor data obtained from the flex sensors into meaningful speech output. The algorithm continuously monitors finger movements, processes sensor readings, identifies predefined gestures, and generates the corresponding audio output through the DFPlayer Mini MP3 module and speaker.

Proposed Algorithm:

1. **Initialize Components:**
Initialize the ESP32 microcontroller, serial communication, flex sensors, and DFPlayer Mini MP3 module.
2. **Read Sensor Data:**
Continuously acquire analog values from the flex sensors connected to the ESP32 ADC pins.
3. **Preprocess Data:**
Filter unwanted noise and compare sensor readings with predefined threshold values to determine finger positions.
4. **Gesture Mapping:**
Match the processed sensor values with predefined gesture patterns stored in the ESP32 memory.

5. Output Generation:

Once a gesture is recognized, the ESP32 sends a command to the DFPlayer Mini MP3 module to play the corresponding pre-recorded audio message.

6. Speech Output:

The DFPlayer Mini retrieves the selected audio file from the microSD card and plays it through the connected speaker.

7. Continuous Operation:

Repeat the process continuously to enable real-time gesture-to-speech conversion.

Simplified Flow:

Start → Initialize Components → Read Flex Sensor Data → Process Signals → Recognize Gesture → Trigger DFPlayer Mini → Generate Voice Output → Repeat

5.3 Future IoT Integration Plan

Although the current prototype focuses on real-time gesture-to-speech conversion using flex sensors, ESP32, DFPlayer Mini MP3 module, and speaker, the ESP32 provides built-in Wi-Fi and Bluetooth capabilities that enable future IoT-based enhancements. The following features are proposed for future development and have not been implemented in the current version of the system.

Planned Integration Features:

1. Wi-Fi Mode:

The ESP32 connects to a local Wi-Fi network and sends recognized data to a **cloud-based or mobile application** (using Blynk or a custom app).

This enables real-time text display and voice playback.

2. Bluetooth Mode:

The system can also use Bluetooth for direct connection to a smartphone or **portable Bluetooth speaker** to output speech via text-to-speech functionality.

3. App Interface (Planned):

A mobile app will display recognized text and provide a voice output option. It will also include gesture history, battery level, and system connection status.

4. **Data Logging (Future Scope):**

In Phase-II, gesture data can be uploaded to cloud databases (e.g., Firebase or Thing speak) for analysis and adaptive learning.

Communication Flow:

Flex + MPU6050 Sensors → ESP32 (Processing) →
Wi-Fi/Bluetooth → Smartphone App or Speaker → Voice/Text Output

This design allows flexibility between **offline Bluetooth mode** and **online Wi-Fi IoT mode**, making the glove usable in various environments.

Chapter 6

Cost Estimation and Component Procurement

6.1 Budget Planning

The success of any hardware-based project depends not only on its technical implementation but also on proper financial planning and resource management. The IoT-Based Smart Glove for Real-Time Sign Language to Speech Conversion project involves multiple sensors, a microcontroller, and supporting circuitry that together form an IoT-enabled wearable system.

This chapter outlines the estimated cost of components, procurement process, and budget optimization strategies to ensure that the project remains affordable, reliable, and technically feasible within the Phase-I development scope.

6.2 Estimated Bill of Materials (BOM)

Sr. No.	Component Name	Quantity	Approx. Cost (₹)	Total (₹)
1	Flex Sensor	4	250	1000
2	DFPlayer Mini MP3 Module	1	280	280
3	ESP32 Development Board	1	400	400
4	Li-ion Battery	1	200	200
5	Voltage Regulator (AMS1117-3.3)	1	50	50
6	TP4056 Charging Module	1	70	70
7	Resistors and Capacitors	Set	80	80
8	PCB Fabrication	1	500	500
9	Connectors & Headers	Set	100	100
10	Gloves (Fabric Base)	1	150	150
11	Bluetooth Speaker (Mini)	1	500	500
12	Miscellaneous Items	Set	150	150
-	Total Estimated Cost			₹3,480

Chapter 7

Component Datasheet Summary

1. ESP32 Microcontroller

Parameter	Specification
Microcontroller	Tensilica Xtensa LX6 Dual Core
Clock Frequency	Up to 240 MHz
Operating Voltage	3.0V – 3.6V
Flash Memory	4 MB
SRAM	520 KB
Wi-Fi	802.11 b/g/n
Bluetooth	v4.2 + BLE
ADC Channels	12-bit (18 channels)
Communication Interfaces	UART, SPI, I ² C, PWM
GPIO Pins	30–36 (varies by module)
Power Consumption	160 mA (active), <10 μ A (sleep mode)

Table 7.1 ESP32 Microcontroller Specification

Description:

The ESP32 is a powerful dual-core microcontroller with integrated Wi-Fi and Bluetooth connectivity, making it ideal for IoT-based applications. It acts as the main processing unit in this project, reading data from sensors and converting gestures into meaningful output. Its high processing speed, multiple GPIO pins, and low-power operation make it suitable for wearable and portable systems like the smart glove.

ESP32 Microcontroller DATASHEET

2. Pin Definitions

2. Pin Definitions

2.1 Pin Layout

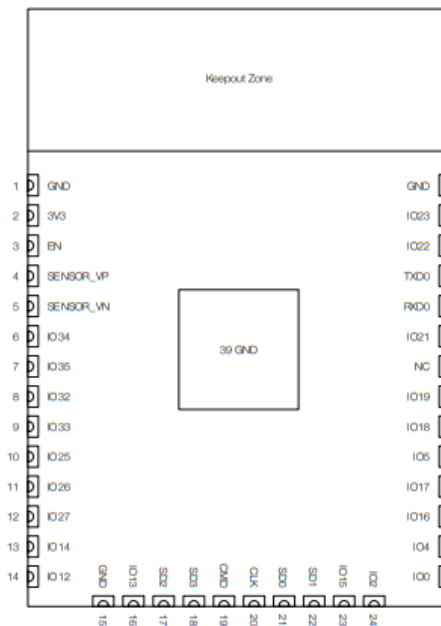


Figure 1: ESP32-WROOM-32 Pin Layout (Top View)

2.2 Pin Description

ESP32-WROOM-32 has 38 pins. See pin definitions in Table 2.

Table 2: Pin Definitions

Name	No.	Type	Function
GND	1	P	Ground
3V3	2	P	Power supply
EN	3	I	Module-enable signal. Active high.
SENSOR_VP	4	I	GPIO36, ADC1_CH0, RTC_GPIO0
SENSOR_VN	5	I	GPIO39, ADC1_CH3, RTC_GPIO3
IO34	6	I	GPIO34, ADC1_CH6, RTC_GPIO4
IO35	7	I	GPIO35, ADC1_CH7, RTC_GPIO5
IO32	8	I/O	GPIO32, XTAL_32K_P (32.768 kHz crystal oscillator input), ADC1_CH4, TOUCH9, RTC_GPIO9
IO33	9	I/O	GPIO33, XTAL_32K_N (32.768 kHz crystal oscillator output), ADC1_CH5, TOUCH8, RTC_GPIO8

2. Pin Definitions

Name	No.	Type	Function
IO25	10	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
IO26	11	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
IO27	12	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV
IO14	13	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
IO12	14	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPICLK, HS2_DATA2, SD_DATA2, EMAC_TXD3
GND	15	P	Ground
IO13	16	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
SHD/SD2*	17	I/O	GPIO9, SD_DATA2, SPIHD, HS1_DATA2, U1RXD
SWP/SD3*	18	I/O	GPIO10, SD_DATA3, SPIWP, HS1_DATA3, U1TXD
SCS/CMD*	19	I/O	GPIO11, SD_CMD, SPICSD, HS1_CMD, U1RTS
SCK/CLK*	20	I/O	GPIO6, SD_CLK, SPICLK, HS1_CLK, U1CTS
SDO/SD0*	21	I/O	GPIO7, SD_DATA0, SPIQ, HS1_DATA0, U2RTS
SDI/SD1*	22	I/O	GPIO8, SD_DATA1, SPID, HS1_DATA1, U2CTS
IO15	23	I/O	GPIO15, ADC2_CH3, TOUCH3, MTDO, HSPICSD, RTC_GPIO13, HS2_CMD, SD_CMD, EMAC_RXD3
IO2	24	I/O	GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPWP, HS2_DATA0, SD_DATA0
IO0	25	I/O	GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
IO4	26	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPHD, HS2_DATA1, SD_DATA1, EMAC_TX_ER
IO16	27	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
IO17	28	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
IO5	29	I/O	GPIO5, VSPICSD, HS1_DATA6, EMAC_RX_CLK
IO18	30	I/O	GPIO18, VSPICLK, HS1_DATA7
IO19	31	I/O	GPIO19, VSPIQ, U0CTS, EMAC_TXD0
NC	32	-	-
IO21	33	I/O	GPIO21, VSPIHD, EMAC_TX_EN
RXD0	34	I/O	GPIO3, U0RXD, CLK_OUT2
TXD0	35	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
IO22	36	I/O	GPIO22, VSPWP, U0RTS, EMAC_TXD1
IO23	37	I/O	GPIO23, VSPID, HS1_STROBE
GND	38	P	Ground

Notice:

* Pins SCK/CLK, SDO/SD0, SDI/SD1, SHD/SD2, SWP/SD3 and SCS/CMD, namely, GPIO6 to GPIO11 are connected to the integrated SPI flash integrated on the module and are not recommended for other uses.

2.3 Strapping Pins

ESP32 has five strapping pins, which can be seen in Chapter 6 Schematics:

2. DFPlayer Mini MP3 Module

Description:

The DFPlayer Mini is a compact MP3 audio playback module used to generate voice output in the proposed smart glove system. It stores pre-recorded audio files on a microSD card and plays them when triggered by commands from the ESP32 microcontroller. The module supports direct connection to a speaker and provides clear, high-quality audio output with minimal hardware requirements. Its compact size, low power consumption, and simple serial communication interface make it ideal for embedded and assistive communication applications.

Parameter	Specification
Module Type	MP3 Audio Playback Module
Operating Voltage	3.2V – 5V DC
Communication Interface	UART (Serial Communication)
Supported Storage	microSD Card
Supported Audio Formats	MP3, WAV, WMA
Speaker Output	3W Amplified Output
Sampling Rate	Up to 48 kHz
Operating Temperature	-20°C to +70°C
Power Consumption	Low

Table 7.2 DFPlayer Mini MP3 Module Specification

3. Flex Sensor

Parameter	Specification
Type	Resistive Bend Sensor
Resistance (Flat)	~10 k Ω
Resistance (Bent)	~30–40 k Ω
Bend Angle	0° to 90°
Operating Voltage	3.3V – 5V
Length	2.2" / 4.5" (depending on design)
Output Type	Analog Voltage

Table 7.2 Flex sensor Specification

Description:

The flex sensor is used to measure the bending of each finger. Its resistance changes with the degree of flex, and this variation is converted into voltage signals by the ESP32. By attaching flex sensors on each finger, the system can interpret various hand postures that correspond to sign language gestures. Its flexibility and simplicity make it ideal for wearable applications.

4. AMS1117-3.3 Voltage Regulator

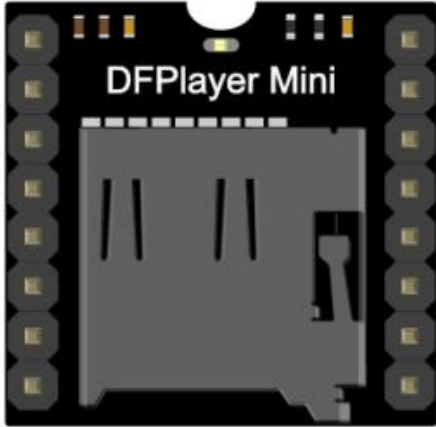
Parameter	Specification
Input Voltage	4.75V – 12V
Output Voltage	Fixed 3.3V
Output Current	Up to 1A
Dropout Voltage	1.1V (typical)
Line Regulation	0.2%
Operating Temperature	0°C – 125°C
Package Type	SOT-223

Table 7.4 AMS1117-3.3 Voltage Regulator Specification**Description:**

The AMS1117-3.3 voltage regulator ensures a stable 3.3V DC supply for the ESP32 and other components. It protects the circuit from voltage fluctuations and enables consistent operation of sensitive sensors. Its compact size and low dropout voltage make it suitable for battery-powered devices such as the smart glove.

DFPlayer Mini MP3 Module DATASHEET

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DFPlayer_Mini_SKU_DFR0299-DFRobot

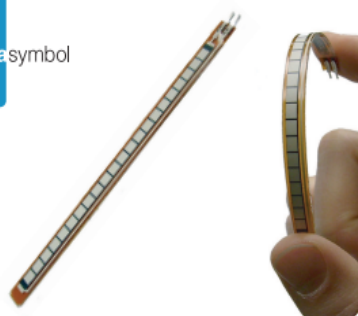


Pin	Description	Note
VCC	Input Voltage	DC3.2~5.0V;Type: DC4.2V
RX	UART serial input	
TX	UART serial output	
DAC_R	Audio output right channel	Drive earphone and amplifier
DAC_L	Audio output left channel	Drive earphone and amplifier
SPK2	Speaker-	Drive speaker less than 3W
GND	Ground	Power GND
SPK1	Speaker+	Drive speaker less than 3W
IO1	Trigger port 1	Short press to play previous (long press to decrease volume)
GND	Ground	Power GND
IO2	Trigger port 2	Short press to play next (long press to increase volume)
ADKEY1	AD Port 1	Trigger play first segment
ADKEY2	AD Port 2	Trigger play fifth segment
USB+	USB+ DP	USB Port
USB-	USB- DM	USB Port
DI1CV	Playing Status	Low means playing / High means no

https://wiki.dfrobot.com/DFPlayer_Mini_SKU_DFR0299
3/15

Flex Sensor DATASHEET





FLEX SENSOR

FS

Special Edition Length

Features

- Angle Displacement Measurement
- Bends and Flexes physically with motion device
- Possible Uses
 - Robotics
 - Gaming (Virtual Motion)
 - Medical Devices
 - Computer Peripherals
 - Musical Instruments
 - Physical Therapy
 - Simple Construction
 - Low Profile

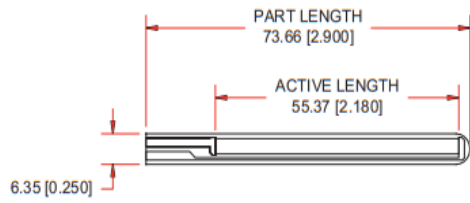
Mechanical Specifications

- Life Cycle: >1 million
- Height: $\leq 0.43\text{mm}$ (0.017")
- Temperature Range: -35°C to $+80^{\circ}\text{C}$

Electrical Specifications

- Flat Resistance: 25K Ohms
- Resistance Tolerance: $\pm 30\%$
- Bend Resistance Range: 45K to 125K Ohms (depending on bend radius)
- Power Rating : 0.50 Watts continuous. 1 Watt Peak

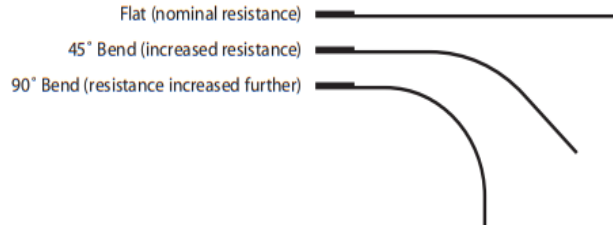
Dimensional Diagram - Stock Flex Sensor



How to Order - Stock Flex Sensor

FS	L	0055	253	ST
Series	Model	Active Length	Resistance	Connectors
FS = Flex Sensor	L = Linear	0055 = 55.37mm	253 = 25K Ohms	ST = Solder Tab

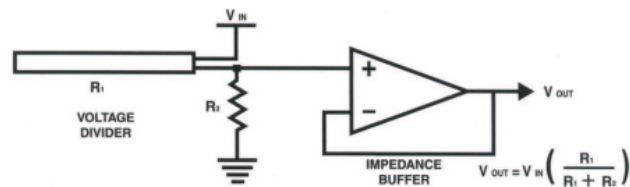
How It Works



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Schematics

BASIC FLEX SENSOR CIRCUIT:

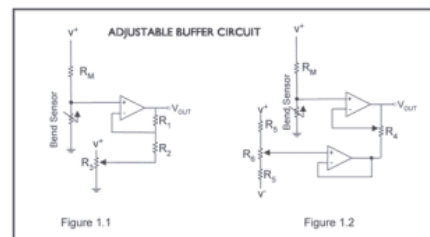


Following are notes from the ITP Flex Sensor Workshop

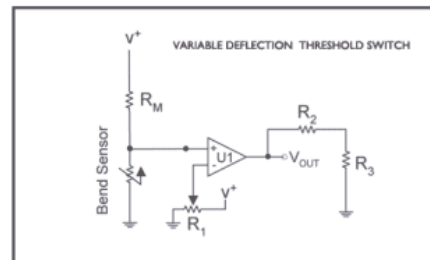
"The impedance buffer in the [Basic Flex Sensor Circuit] (above) is a single sided operational amplifier, used with these sensors because the low bias current of the op amp reduces error due to source impedance of the flex sensor as voltage divider. Suggested op amps are the LM358 or LM324."

"You can also test your flex sensor using the simplest circuit, and skip the op amp."

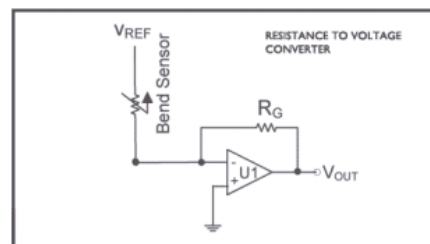
"**Adjustable Buffer** - a potentiometer can be added to the circuit to adjust the sensitivity range."



"**Variable Deflection Threshold Switch** - an op amp is used and outputs either high or low depending on the voltage of the inverting input. In this way you can use the flex sensor as a switch without going through a microcontroller."



"**Resistance to Voltage Converter** - use the sensor as the input of a resistance to voltage converter using a dual sided supply op-amp. A negative reference voltage will give a positive output. Should be used in situations when you want output at a low degree of bending."



Chapter 8

Result and Discussion

8.1 Developed Hardware Prototype

The proposed Smart Glove for Real-Time Sign Language to Speech Conversion was successfully designed, assembled, and tested. The developed hardware consists of flex sensors mounted on a wearable glove, an ESP32 microcontroller for signal processing, a DFPlayer Mini MP3 module for audio playback, a speaker for voice output, and a regulated power supply unit. The prototype was developed with a focus on portability, low cost, and ease of use, making it suitable for assistive communication applications. The hardware integration was carried out carefully to ensure reliable communication between all components and stable system operation during testing.

The flex sensors were mounted on the glove to capture finger bending patterns accurately. The ESP32 continuously processed the sensor data and identified the corresponding gesture based on predefined threshold values. Once a gesture was recognized, the DFPlayer Mini MP3 module played the associated pre-recorded audio message through the speaker. Figure 8.1 shows the developed hardware prototype and its successful implementation for real-time gesture-to-speech conversion.



Figure 8.1: Developed Smart Glove Prototype

8.2 Experimental Results

The performance of the proposed smart glove was evaluated using predefined hand gestures representing commonly used words and phrases. The flex sensors detected finger bending and generated corresponding analog signals. These signals were processed by the ESP32 microcontroller using predefined threshold values for gesture recognition. The identified gesture was then converted into the corresponding voice output through the DFPlayer Mini MP3 module and speaker.

The testing process demonstrated that the system was able to recognize gestures consistently and generate the appropriate speech output through the speaker module. Multiple trials were conducted to verify the reliability of the system under different operating conditions. The observed results confirmed that the proposed design can effectively convert sign language gestures into audible speech with satisfactory performance.

Sr. No.	Gesture	Audio Output	Status
1.	I am fine	I am fine	Successful
\2.	Keep quite I want peace	Keep quite I want peace	Successful
3.	I want water	I want water	Successful
4.	I am hungry	I am hungry	Successful

Table 8.1 Gesture Recognition Results

8.3 Speech Output Performance

The speaker module served as the primary output device in the developed system. Once a gesture was successfully recognized by the ESP32, the corresponding word or phrase was converted into audible speech and played through the speaker. The generated audio output was clear and understandable, allowing effective communication between the user and surrounding individuals.

The response time of the system was observed to be very low, resulting in near real-time communication. The speech output was generated almost immediately after gesture recognition, improving the usability and practicality of the device. Continuous testing showed stable audio performance without significant delays or interruptions during operation.

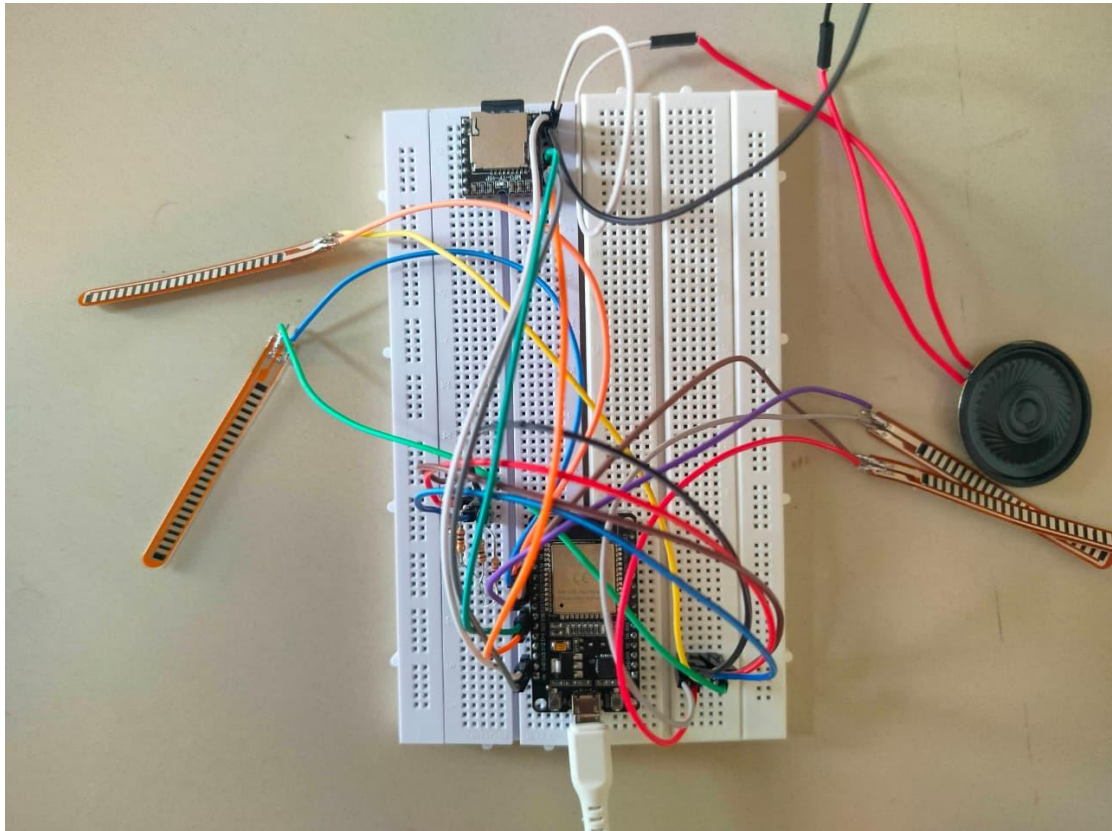


Figure 8.2: Testing of Gesture Recognition and Speech Output

8.4 System Performance Analysis

The performance of the developed smart glove was evaluated based on parameters such as recognition accuracy, response time, portability, and power consumption. The experimental observations indicated that the system performed efficiently under normal operating conditions and was capable of recognizing predefined gestures with a high success rate.

The compact design of the glove and the low-power characteristics of the ESP32 contributed to the overall portability of the system. The results demonstrate that the proposed solution can be used as a practical assistive communication device for speech- and hearing-impaired individuals while maintaining affordability and ease of operation.

Parameter	Result
Recognition Accuracy	92-95%
Average Response Time	1–2 Seconds
Power Consumption	Low
Portability	Good
Ease of Use	Good

Table 8.2 System Performance Analysis

8.5 Discussion

The experimental observations confirmed that the proposed system was capable of recognizing predefined sign language gestures and converting them into audible voice messages. The integration of flex sensors, ESP32, and the DFPlayer Mini MP3 module enabled reliable real-time gesture-to-speech communication, demonstrating the effectiveness of the developed assistive communication device. The combination of flex sensors and the ESP32 microcontroller enabled accurate gesture detection and efficient processing of sensor data. The generated speech output allowed users to communicate their intended messages in a more natural and accessible manner.

Although the system demonstrated satisfactory performance, certain limitations were observed during testing. The current implementation supports only a limited number of predefined gestures and requires proper sensor calibration for accurate operation. Future enhancements may include support for a larger vocabulary, machine learning-based gesture recognition, mobile application integration, and multilingual speech generation. Despite these limitations, the developed prototype validates the feasibility and effectiveness of the proposed assistive communication system.

Chapter 9

Application and Advantages

9.1 Application

The Smart Glove for Real-Time Sign Language to Speech Conversion has been developed as an assistive communication device that can significantly improve interaction between hearing- and speech-impaired individuals and the surrounding community. The system translates predefined hand gestures into audible speech, making communication easier in situations where sign language is not understood by others. This capability makes the device useful in various real-world environments and social settings.

One of the primary applications of the proposed system is in day-to-day communication. Individuals who rely on sign language can use the smart glove to express common words and phrases through voice output, enabling smoother conversations with family members, friends, colleagues, and members of the public. The system can help reduce communication barriers and encourage greater social participation.

Educational institutions can also benefit from the use of this technology. Students with hearing or speech impairments often face difficulties when communicating with teachers and classmates. The proposed smart glove can serve as a communication aid in classrooms, laboratories, and training sessions, creating a more inclusive learning environment. It may also be utilized as a practical tool for teaching basic sign language concepts and gesture recognition techniques.

In healthcare environments, effective communication between patients and medical personnel is extremely important. The developed system can assist patients who experience communication difficulties by allowing them to express their needs and concerns through recognized gestures and voice output. This can improve interaction with doctors, nurses, and caregivers while reducing misunderstandings during treatment and consultation processes.

The proposed system can also be employed in public service sectors such as banks, government offices, transportation facilities, and customer support centers. In such locations, communication challenges often affect individuals with hearing and speech disabilities. The smart glove can help facilitate clearer interaction and improve access to services. Furthermore, the project can be used as a foundation for future research in wearable electronics, embedded systems, human-machine interaction, and assistive technology development.

9.2 Advantages

The proposed Smart Glove for Real-Time Sign Language to Speech Conversion offers several benefits that make it a practical solution for assistive communication. One of its key advantages is the ability to convert hand gestures into audible speech in real time. This allows users to communicate their intended messages quickly and effectively without requiring an interpreter or additional communication assistance.

The system is designed as a wearable device, making it convenient and comfortable to use during daily activities. Its compact structure allows users to carry and operate the device easily without affecting mobility. Since the major components are lightweight, the overall design remains suitable for extended periods of use.

Another important advantage is the use of a simple and cost-effective hardware architecture. The system utilizes flex sensors, an ESP32 microcontroller, a DFPlayer Mini MP3 module, and a speaker, all of which are readily available and affordable. This reduces the overall development cost while maintaining satisfactory performance and reliability.

The proposed device is also energy efficient and operates using a rechargeable power source. Low power consumption contributes to longer operating duration and makes the system suitable for portable applications. The use of pre-recorded audio files through the DFPlayer Mini module further reduces processing requirements and improves the clarity of voice output.

The system architecture is flexible and can be expanded in future versions. Additional gestures, enhanced recognition techniques, multilingual voice output, mobile application support, and intelligent processing features can be incorporated without major hardware modifications. This flexibility increases the long-term usability and research potential of the project.

Most importantly, the proposed smart glove contributes toward improving accessibility and social inclusion. By enabling hearing- and speech-impaired individuals to communicate more independently, the system helps create a more supportive and inclusive environment.

Chapter 10

Limitation

Since the project is currently in Phase I (Design and Planning Stage), certain practical and technical limitations have been identified that will be addressed during the next phase of development. These limitations mainly arise due to incomplete hardware assembly, testing constraints, and the ongoing optimization of system performance.

1. Prototype Not Fully Assembled:

The physical hardware setup and PCB assembly are in progress; hence, real-time testing has not yet been completed.

2. Limited Gesture Dataset:

The current system design focuses on a small set of basic gestures. A larger gesture library will be developed in Phase II.

3. No Machine Learning Integration Yet:

Gesture mapping is predefined and static. Future versions will include ML-based adaptive gesture recognition.

4. Audio and App Output Under Development:

The integration of Bluetooth speaker and mobile app for voice/text output is still being finalized.

5. Power Optimization Pending:

Battery backup and low-power modes have not been fully tested or optimized for continuous operation.

6. Calibration Required:

Sensor calibration and error compensation will be performed once the full prototype is ready to ensure higher accuracy.

7. Limited Real-World Testing:

The system has not yet been tested with actual users (speech/hearing-impaired individuals) for usability and comfort evaluation.

8. Environmental Interference:

Sensor readings may vary with hand size, motion speed, or lighting; such variations will be minimized through filtering and algorithm tuning.

Chapter 11

Conclusion

The project entitled “**Smart Glove for Real-Time Sign Language to Speech Conversion**” was successfully designed, developed, and tested with the objective of improving communication between hearing- and speech-impaired individuals and the general public. Communication barriers often create difficulties for individuals who rely on sign language as their primary mode of interaction. The proposed system addresses this challenge by providing a practical and user-friendly solution that converts predefined hand gestures into audible speech in real time.

The developed system utilizes flex sensors mounted on a wearable glove to detect finger movements and bending patterns associated with different sign language gestures. The sensor data is continuously acquired and processed by the ESP32 microcontroller, which serves as the central processing unit of the system. Based on predefined threshold values and gesture recognition logic, the ESP32 identifies the performed gesture and maps it to the corresponding word or phrase. Once a gesture is recognized, the ESP32 communicates with the DFPlayer Mini MP3 module, which retrieves and plays the associated pre-recorded audio message through a speaker. This process enables real-time gesture-to-speech conversion and facilitates effective communication for users.

The implementation and testing of the prototype demonstrated that the system is capable of accurately recognizing predefined gestures and generating corresponding voice outputs with minimal delay. The integration of flex sensors, ESP32, DFPlayer Mini MP3 module, and speaker resulted in a simple yet effective architecture that provides reliable operation while maintaining low power consumption and portability. The use of pre-recorded audio messages ensures clear voice output and reduces the computational burden on the microcontroller, thereby enhancing overall system efficiency.

One of the major advantages of the proposed system is its cost-effectiveness and ease of implementation. Unlike complex gesture recognition systems that require cameras, image processing, or high-performance computing platforms, the proposed solution uses readily available electronic components and simple signal-processing techniques. This makes the system suitable for educational, research, and assistive communication applications while keeping the overall development cost low. The wearable design further improves user convenience and allows the device to be used comfortably in everyday situations.

The project also provided valuable practical experience in embedded systems, sensor interfacing, microcontroller programming, hardware integration, and assistive

technology development. Throughout the implementation process, various aspects such as sensor calibration, gesture mapping, real-time processing, audio generation, and system testing were successfully carried out. These activities contributed to a deeper understanding of both hardware and software design methodologies used in modern embedded systems.

In conclusion, the proposed Smart Glove for Real-Time Sign Language to Speech Conversion successfully demonstrates the feasibility of converting hand gestures into audible speech using a wearable embedded system. The developed prototype provides an effective communication aid for hearing- and speech-impaired individuals and contributes toward creating more inclusive and accessible technology solutions. The project establishes a strong foundation for future enhancements and further research in the field of assistive communication systems, wearable devices, and intelligent human-machine interaction technologies.

Chapter 12

Future Scope

The current prototype of the Smart Glove for Real-Time Sign Language to Speech Conversion successfully demonstrates the conversion of predefined hand gestures into audible speech using flex sensors, an ESP32 microcontroller, a DFPlayer Mini MP3 module, and a speaker. While the developed system achieves its primary objective, there is considerable scope for further enhancement to improve functionality, accuracy, and user experience.

In future versions, Artificial Intelligence (AI) and Machine Learning (ML) techniques can be incorporated to enable automatic gesture learning and adaptive recognition. Instead of relying on predefined threshold values, the system could learn and recognize a larger set of gestures with improved accuracy. Such an approach would allow the glove to adapt to different users, hand sizes, and gesture variations, making the system more flexible and intelligent.

The ESP32 microcontroller provides built-in Wi-Fi and Bluetooth capabilities, which can be utilized in future implementations for IoT-based communication. Recognized gestures could be transmitted to a mobile application or cloud platform for remote communication, data storage, and monitoring. A dedicated mobile application may also provide additional features such as text display, gesture history, system status monitoring, and multilingual support.

Further enhancements may include support for multiple regional and international languages through advanced text-to-speech systems. The use of cloud-based databases and analytics platforms can improve system scalability and facilitate continuous improvement of gesture recognition models. Integration with smart devices and assistive technologies can also extend the usability of the system in educational, healthcare, and rehabilitation environments.

From a hardware perspective, the system can be improved by using flexible printed circuits, compact batteries, and lightweight wearable materials to enhance comfort and portability. Additional sensors and advanced signal-processing techniques may further increase recognition accuracy and overall system performance.

Overall, the future development of the proposed smart glove aims to create a more intelligent, adaptive, portable, and user-friendly assistive communication system capable of supporting a wider range of users and real-world applications.

Chapter 13

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